Executive Summary

Renumbering Tracks for a Multidisc Compilation Album

An Advanced SQL Window Function Analysis Using the Chinook Database

Prepared For: Record Label Management

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1. Purpose

Management asked how the tracks should be renumbered if a new multidisc album were created with artists from a similar genre. Renumbering means giving every track a position on the album and, because the album has more than one disc, showing which disc it belongs to and where it falls on that disc. Doing this by hand is slow and error-prone, so this analysis uses SQL window functions to build the sequence automatically from the Chinook music database.

2. Approach and SQL Implementation

The example album uses six Rock artists: Led Zeppelin, Deep Purple, Pearl Jam, Van Halen, Santana, and Foo Fighters. The query works in stages, each of which uses a window function, a calculation that looks across a group of rows without collapsing them:

- **Choosing the tracks.** The query selects the Rock tracks of the six artists. It leaves out tracks longer than ten minutes and one damaged record whose title field contained thousands of characters of unrelated data.
- **Removing repeats.** Some songs exist in the database in several versions. ROW_NUMBER, grouped by artist and title, keeps only the longest version of each song.
- **Picking each artist's best-length tracks.** A second ROW_NUMBER, grouped by artist, keeps each artist's four longest remaining tracks.
- **Mixing the artists.** Tracks are ordered so that each artist appears in turn instead of one artist filling a whole disc.
- **Splitting into discs.** SUM used as a running total adds up playing time as the album is built, and a new disc starts after every 80 minutes.
- **Renumbering.** A final ROW_NUMBER, restarted for each disc with PARTITION BY, gives every track its number on its disc, alongside its overall position on the album.

Advanced SQL Query Used

```
WITH selected_tracks AS (
    SELECT
        ar.name AS artist_name,
        t.name AS track_name,
        t.milliseconds::numeric AS ms
    FROM chinook.track t
    JOIN chinook.genre g ON t.genreid::integer = g.genreid::integer
    JOIN chinook.album al ON t.albumid::integer = al.albumid::integer
    JOIN chinook.artist ar ON al.artistid::integer = ar.artistid::integer
    WHERE g.name = 'Rock'
      AND ar.name IN ('Led Zeppelin', 'Deep Purple', 'Pearl Jam',
                      'Van Halen', 'Santana', 'Foo Fighters')
      AND LENGTH(t.name) < 100
      AND t.milliseconds::numeric <= 600000
),
one_version AS (
    SELECT
        artist_name, track_name, ms,
        ROW_NUMBER() OVER (PARTITION BY artist_name, LOWER(track_name)
```

```
                            ORDER BY ms DESC) AS version_rank
    FROM selected_tracks
),
top_per_artist AS (
    SELECT
        artist_name, track_name, ms,
        ROW_NUMBER() OVER (PARTITION BY artist_name ORDER BY ms DESC) AS artist_rank
    FROM one_version
    WHERE version_rank = 1
),
album_tracks AS (
    SELECT * FROM top_per_artist WHERE artist_rank <= 4
),
sequenced AS (
    SELECT
        artist_name, track_name, ms,
        ROW_NUMBER() OVER (ORDER BY artist_rank, artist_name) AS overall_track_number,
        SUM(ms) OVER (ORDER BY artist_rank, artist_name
                      ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) - ms AS start_ms
    FROM album_tracks
)
SELECT
    FLOOR(start_ms / 4800000)::int + 1 AS disc_number,
    ROW_NUMBER() OVER (PARTITION BY FLOOR(start_ms / 4800000)
                       ORDER BY overall_track_number) AS track_on_disc,
    overall_track_number,
    artist_name,
    track_name,
    ROUND(ms / 60000, 2) AS minutes
FROM sequenced
ORDER BY overall_track_number;
```

Query Results

The table below shows the new album. "Track" is the number on that disc, and "Overall" is the position across the whole album.

Disc	Track	Overall	Artist	Track Title	Minutes
1	1	1	Deep Purple	Burn	7.57
1	2	2	Foo Fighters	Walking After You	5.06
1	3	3	Led Zeppelin	Tea For One	9.45
1	4	4	Pearl Jam	Release	9.10
1	5	5	Santana	Jingo	9.88
1	6	6	Van Halen	Year to the Day	8.58
1	7	7	Deep Purple	Lazy	7.37
1	8	8	Foo Fighters	Everlong	4.18
1	9	9	Led Zeppelin	Stairway To Heaven	8.83
1	10	10	Pearl Jam	Black	6.93
1	11	11	Santana	Evil Ways	7.92
2	1	12	Van Halen	Once	7.71
2	2	13	Deep Purple	Anya	6.54
2	3	14	Foo Fighters	Enough Space	2.62
2	4	15	Led Zeppelin	In The Light	8.78
2	5	16	Pearl Jam	Daughter	6.79
2	6	17	Santana	Primavera	6.31
2	7	18	Van Halen	Without You	6.50
2	8	19	Deep Purple	This Time Around / Owed to 'G' [Instrumental]	6.17
2	9	20	Led Zeppelin	Kashmir	8.48
2	10	21	Pearl Jam	F*Ckin' Up	6.29
2	11	22	Santana	Do You Like The Way	5.91
2	12	23	Van Halen	How Many Say I	6.07

The query produced a 23-track album of about 163 minutes over two discs. Disc 1 has 11 tracks and Disc 2 has 12. The track number starts again at 1 on Disc 2, while the overall number runs continuously from 1 to 23. Each artist appears in turn, so the album moves from one act to the next instead of grouping them together. Foo Fighters contributes three tracks because only three of its Rock tracks met the length rules, and each of the other artists contributes four. Disc 1 runs about 85 minutes, slightly over the 80-minute target, because a track stays on the disc where it starts; this is discussed in the limitations below.

3. Defensible Position and Evaluation

Three Reasons Why This Is a Good Approach

1. **It is automatic and repeatable.** Because the numbering comes from window functions, changing the list of artists or the length rules and running the query again renumbers the whole album instantly. There is no manual editing, so numbering mistakes are avoided.
2. **It solves the multidisc problem directly.** The running total decides where each disc begins, and PARTITION BY restarts the numbering on each disc. The result gives management both the disc-and-track label used on the packaging and the overall album position.
3. **It produces a balanced album.** Keeping a single version of each song avoids repeats, and rotating through the artists gives the listener variety while the shared genre keeps the collection consistent. No single artist dominates a disc.

Three Limitations of the Approach

1. **It does not judge musical flow or popularity.** Tracks are chosen and ordered by length and artist rotation, not by how well one song leads into the next or by how well known it is. A well-loved hit could be left out because it is longer than ten minutes or is not among an artist's four longest tracks.
2. **The disc limit is approximate.** A track stays on the disc where it starts, so Disc 1 runs to about 85 minutes, above a strict 80-minute limit. A hard cap would need extra logic that moves any track that does not fit onto the next disc.
3. **It depends on the quality of the data and on one genre label.** One record in the database was corrupted and had to be filtered out, and several songs appear in more than one version. Each track also carries only one genre label, so artists whose music crosses genres can be missed.

What Additional Data Would Improve This Approach

Sales figures or streaming counts would let the query select the best-known songs instead of the longest ones. Audio features such as tempo and energy would allow the tracks to be ordered so that each song flows into the next. Release dates and licensing terms would show whether each track can legally and affordably be included, and the exact capacity of the intended disc format would make the disc split precise.

4. Conclusion

Using SQL window functions, a multidisc compilation can be renumbered automatically: the query removes duplicate songs, mixes the artists, splits the tracks into discs, and restarts the numbering on each one. The example album of 23 tracks over two discs shows the method working. It is a strong, repeatable foundation, and combining it with popularity, flow, and licensing data would let the label plan real releases with more confidence.